Carrera Race Monitor

Assembly Instructions

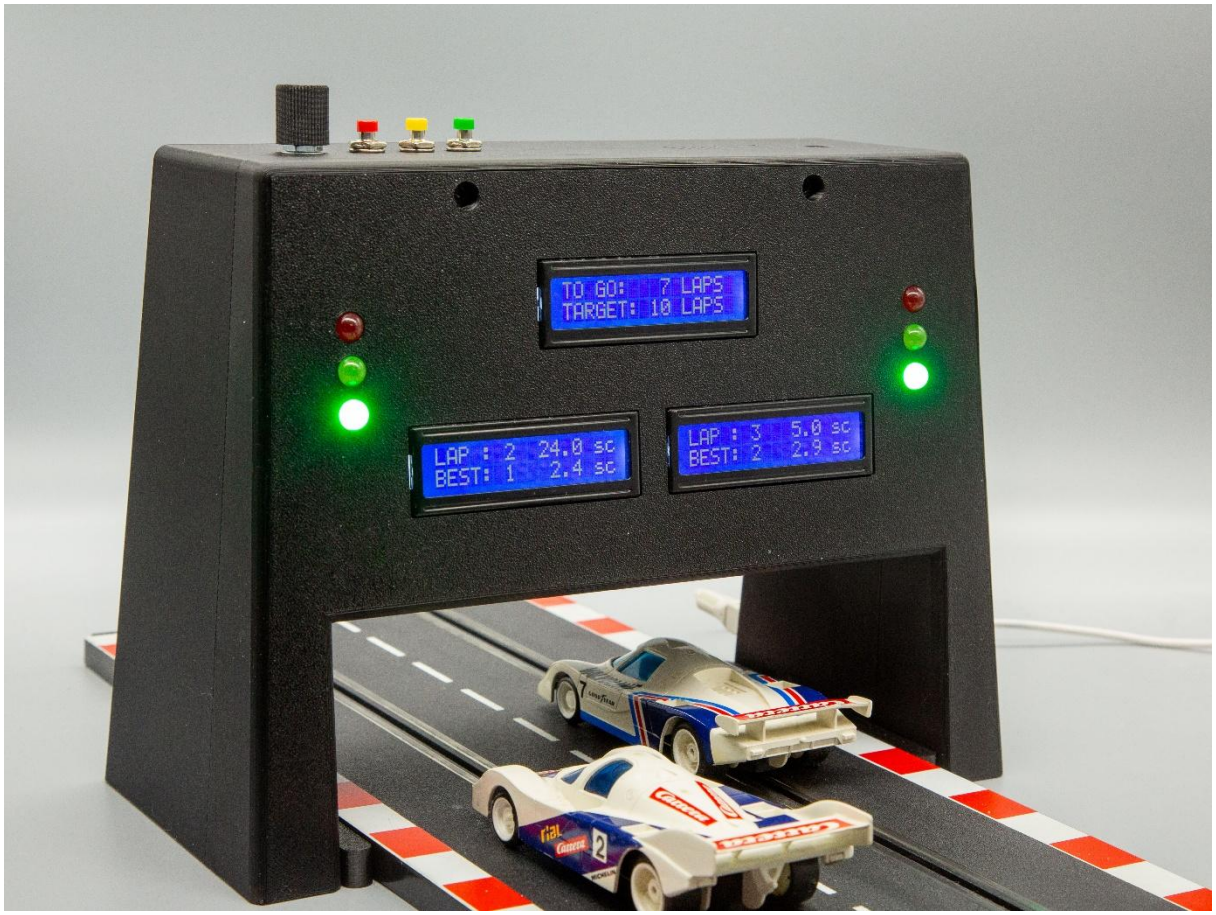


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1 Preparation

1.1 Parts

Part #	Description	Model	Req.	Source	Sales Qnty	Price	Cost	Remark
#01	Raspberry Pi	Zero WH	1	https://amzn.eu/d/0cRGxUfx	1	39,99	39,99 €	§1, §2, §3, §4
#02	Micro SD Card	32 GB or larger	1	https://amzn.eu/d/0cVfw0CE	2	30,99	15,50 €	
#03	Power Supply	5V, 2.5 A	1	https://amzn.eu/d/03NTup31	1	9,49	9,49 €	§5
#04	USB-C to Micro USB Adapter		1	https://amzn.eu/d/0hMWu9zx	3	4,95	1,65 €	
#05	LCD Display	AZDelivery 3 x HD44780 16 x 2 LCD Module Display Bundle with I2C Interface	3	https://amzn.eu/d/07rp2ziM	3	10,09 €	10,09 €	
#06	Logic Level Converter 3.3V 5V	TXS0108E	1	https://amzn.eu/d/002lTpKL	6	8,69 €	1,45 €	
#07	Circuit Board	PCB 160 x 100 mm Strip Grid Board Copper	0,33	https://amzn.eu/d/06ZO410I	1	5,49 €	1,81 €	
#08	Traffic Light	AZDelivery Traffic Light Module	2	https://amzn.eu/d/0eYiQ6vp	5	5,19 €	2,08 €	
#09	Obstacle Sensor	Hailege LM393 3-Wire IR Reflective Photoelectric Sensor	2	https://amzn.eu/d/00Azx0tn	10	9,59 €	1,92 €	
#10	Push Button	7 mm Pre-Wired Mini Momentary Push Button	3	https://amzn.eu/d/00PTFzne	12	7,89 €	1,97 €	
#11	Rotary Encoder	KY-040 Rotary Encoder Module 360° Rotary Encoder with Knob Pack of 5	1	https://amzn.eu/d/02uSWvmZ	5	6,99 €	1,40 €	
#12	Active Buzzer	Active Buzzer Module DC 3.3V-5V Signal Sound Alarm Sensor Module Low Level Trigger Buzzer Control Board	1	https://amzn.eu/d/02J0ZU2s	3	5,49 €	1,83 €	
#13	40-Pin GPIO Rainbow Ribbon Cable	ZDE 40-Pin GPIO Rainbow Ribbon Cable Computer Accessories 20 cm Compatible with Raspberry Pi 5 4B 3 2 Model B B+ (Male to Female)	1	https://amzn.eu/d/02IsTBP4	1	6,99 €	6,99 €	
#14	USB C Cable	USB C 3.1 10Gbps Male to USB C Female Screw Cable for Panel Mounting	1	https://amzn.eu/d/0dHg8eoC	1	11,19 €	11,19 €	
#15	USB-C to Micro USB Adapter	USB-C to Micro USB Adapter [Pack of 3] USB Type C Female to Micro USB Male Quick Charge (Black)	1	https://amzn.eu/d/0dyDGZxG	3	4,95 €	1,65 €	

Remarks:

§1 In the project, a Raspberry Pi Zero 2 WH has been used, but a Raspberry Pi Zero WH will do as well.

§2 Make sure to use a Raspberry Pi with on-board Wireless LAN (W)

§3 If available, use a variant with a 40-pin GPIO Header already pre-soldered (H). The pins must be on the front side of the device.

§4 The sales price for a Raspberry Pi Zero, at the time of writing, was unusually high, which is assumed to be due to a shortage of older cheap components due to currently limited production capacity. The usual price for a Zero W is about 17 €, with a little bit more for the Zero 2 W.

§5 The Raspberry Pi Zero has a Micro USB socket. In operation, a USB-C socket will be used for power supply, but during testing an adapter (#04) must be used.

1.2 Stock Items

The following list shows required items, which may be available in stock, such as various screws.

Item #	Description	Model	Required	Source	Sales Qnty	Price	Cost
#100	JST 2.54 mm XH-5 male connector for circuit boards		1	https://amzn.eu/d/0bRL9lFD	310	7,99 €	0,03 €
#101	JST 2.54 mm XH-4 male connector for circuit boards		3	See #100	310	7,99 €	0,08 €
#102	24 AWG Solid-Core Wire Kit, PVC-Coated	Fermerry	0,5	https://amzn.eu/d/018R1h6N	45	17,29 €	0,19 €
#103	JST XH2.54 Socket 2P male	XH 2.54mm Connector Pin Header & Ribbon Cable Kit Compatible JST XH Male 2/3/4/5/6/7/8 Male Housing	3	https://amzn.eu/d/0dberzft	55	15,99 €	0,87 €
#104	JST XH2.54 Socket 3P male	Included in #103	3	See #103	55	15,99 €	0,87 €
#105	JST XH2.54 Socket 4P male	Included in #103	2	See #103	55	15,99 €	0,58 €
#106	JST XH2.54 Socket 5P male	Included in #103	2	See #103	55	15,99 €	0,58 €
#107	JST XH2.54 Connector 2P female	GTIWUNG 480-Piece JST-XH 2.54 mm Connector Set 2.54 mm Pitch Female Pin Header, 2/3/4/5/6 Pin Housing JST Adapter Cable Male and Female	3	https://amzn.eu/d/0hjBtLqG	480	7,99 €	0,05 €
#108	JST XH2.54 Connector 3P female	Included in #107	3	See #107	480	7,99 €	0,05 €
#109	JST XH2.54 Connector 4P female	Included in #107	5	See #107	480	7,99 €	0,08 €
#110	JST XH2.54 Connector 5P female	Included in #107	3	See #107	480	7,99 €	0,05 €
#111	24 AWG Stranded Wire, Silicone	Fermerry 24 AWG Stranded Wire, Silicone, Tinned Copper Wire Coil, 6 Colours/3 m Each, Flexible 24 Gauge Hook-up Electric Cable Kit, 0.2 mm^2 Cable, Wire, Connection Cable Set, 200 °C, Assortment	4,2	https://amzn.eu/d/0j31d7ef	18	15,19 €	3,54 €
#112	Dupont connector 3P female	Treki 620pcs Dupont Connector Kit 2.54mm Pitch Dupont Connector Housing 1/2/3/4/5/6 Pin Dupont Male Female Pin Header Dupont Crimp Pins Repair Circuit Board	3	https://amzn.eu/d/0aPNh0uM	620	8,99 €	0,04 €
#113	Dupont connector 4P female	Included in #113	5	See #113	620	8,99 €	0,07 €
#114	Dupont connector 5P female	Included in #113	3	See #113	620	8,99 €	0,04 €
#115	Cable labels	Caianwin Pack of 200 Cable Labels 10 Colours Waterproof Cable Marking	20	https://amzn.eu/d/0dijC8ml	200	4,99 €	0,50 €
#116	Heat Shrink Tubing	Eventronic ET1001 Heat Shrink Tubing - Electrical Insulation - Heat Shrink Wrap Cable Sleeves - 5 Colours, 12 Sizes, 560 Pieces	4	https://amzn.eu/d/0aulVSY4	560	7,49 €	0,05 €

Item #	Description	Model	Required	Source	Sales Quantity	Price	Cost
#150	Threaded Insert M4x8x6	360 x Threaded Insert M2 M2.5 M3 M4 M5 M6	2	https://amzn.eu/d/0fQnbrcJ	360	11,89 €	0,07 €
#151	Cylinder Screw M4x16MM	1230 Piece M2 M3 M4 M5 Screws Set, Cylinder Screws with Hexagon Socket	2	https://amzn.eu/d/02jguqhI	1230	10,58 €	0,02 €
#152	Threaded Insert M4x4x6	Included in #150	2	See #150	360	11,89 €	0,07 €
#153	Cylinder Screw M4x8MM	Included in #151	4	See #151	1230	10,58 €	0,03 €
#154	Cylinder Screw M3x6MM	Included in #151	19	See #151	1230	10,58 €	0,16 €
#155	Threaded Insert 1/4"x6.4x8	Short UNC Thread Insert 1/4 Inch Pack of 20 RX-1/4 Inch 20 x 6.4	1	https://amzn.eu/d/050ffsfM	20	9,99 €	0,50 €
#156	USB C Cable	USB C 3.1 10Gbps Male to USB C Female Screw Cable for Panel Mounting	1	https://amzn.eu/d/0dHg8eoC	1	11,19 €	11,19 €
#157	Cylinder Screw M2.5x8	300pcs M2.5 x 3-20mm Hex Socket Cap Screw Set 12.9 Degree Black Alloy Steel Socket Head Cap Screw with Nut	4	https://amzn.eu/d/02TtpPMP	300	9,03 €	0,12 €
#158	Threaded Insert M4x6x6	Included in #150	2	See #150	360	11,89 €	0,07 €
#159	M4 x 12 mm Washers	Pack of 60 M4 x 12 mm Plastic Washers Black	2	https://amzn.eu/d/05JZ5ayJ	60	4,39 €	0,15 €

1.3 Tools

The following list makes suggestions for tools to be used:

Tool #	Description	Model	Required	Source	Sales Quantity	Price	Cost	Remark
#200	Soldering Equipment	Weller WE 1010	1	https://amzn.eu/d/0f4Y4Rm0	1	163,99 €	163,99 €	§1
#201	PCB Clamp	Self-printed	1	Model from makerworld	1		0,00 €	
#202	Fibreglass Eraser	Fibreglass Eraser	1	https://amzn.eu/d/0c6Nza4S	1	10,70 €	10,70 €	
#203	Multimeter							
#204	Crimping Tool	Preciva 2220pc DuPont Crimping Tool Crimping Tool Set Fits 0.08-1.5 (AWG28-16) mm Non-Insulated Cable Lugs and Sockets, Suitable for ATX, EPS, PCIE, SATA Power Pins		https://amzn.eu/d/0852hMcK		36,99 €		
#205	Heat Insert Press	Self-printed		Model from makerworld				
#206	PH1 Screwdriver with long shaft							
#207	Hex Key Allen Wrench 3.5 mm							
#208	Socket Wrench 10 mm							
#209	Metal Drill Bit 8.0 mm							

Remarks:
§1 Any soldering iron, suitable for electronics, will do.

2 Prepare Raspberry Pi Zero

In this step, you will install the Raspberry Pi OS and afterwards the Carrera Race Monitor software.

1. Follow the Raspberry Pi Getting Started Documentation for OS installation using Imager.

   NOTE: It is recommended installing the Debian distribution proposed by Imager for your Raspberry Pi device. This is usually the full 64-bit version including desktop. The latest distribution is currently Trixie.

   Make sure that the Wi-Fi credentials are correctly entered and that SSH is enabled.

2. Once the SD card is written, insert it into the Raspberry Pi Zero and power it up.

   Initially, it will take a while until the device is visible in your network.

3. Connect to the Pi using SSH:

   ```
   ssh <user>@<host>
   ```

   with `<user>` and `<host>` as specified during setup with Imager

4. Make sure that the system is up to date

   ```
   sudo apt update
   ```

   ```
   sudo apt full-upgrade
   ```

5. Run the automatic installer with:

   ```
   bash <(curl -fsSL
   https://raw.githubusercontent.com/signag/Carrera_Race_Monitor/m
   ain/scripts/install_Carrera_Race_Monitor.sh)
   ```

   (For details, see below)

6. As a result, Carrera_Race_Monitor will have been installed at

   ```
   ~/prg/Carrera_Race_Monitor
   ```

   with Python scripts located in the `code` subdirectory

2.1 Automated Installer Steps

2.1.1 Preparation

```
================================================
=== Carrera_Race_Monitor Automated Installer    ===
===                                             ===
=== Exit at any step with Ctrl+C                ===
================================================

RPI Model           : Raspberry Pi Zero 2 W Rev 1.0
Detected OS codename: trixie full
Hostname            : raspi12

Running as user     : sn
Installing at       : /home/sn/prg

=================
Installation Mode
=================
Installation Path     : /home/sn/prg/Carrera_Race_Monitor (will be created)

This is the first installation of Carrera_Race_Monitor on this system.
Services will not be installed to allow manual device tests during system
assembly.
After successful device tests, you can run the installer again to configure
and start the services.

No more questions! Ready to start installation? [Y/n]:
```

At this step, you can continue or abort the installation.

2.1.2 Protocol

The following is a shortened installation protocol showing the different installation steps:

```
Creating root directory ...
Created: /home/sn/prg

Checking git ...
git is installed

Cloning Carrera_Race_Monitor ...
Cloning into 'Carrera_Race_Monitor'...
...

Creating virtual environment ...
Created: .venv

Activating virtual environment ...
Virtual environment activated
(.venv)

Installing Flask ...
...

Installing flask-cors ...
...

Installing gpiozero ...
...
```

```
Installing libraries for LCD display ...
...

Installing I18n tools ...
...

Installing messaging infrastructure ...
...

Checking I2C interface status ...
...

Virtual environment deactivated
(.venv)

Checking Flask service port ...
Trying port 5100 ...
Using port 5100

Services installation skipped.
This allows manual device tests during system assembly.
After assembly and successful device tests rerun this installer to install
the services.
For details refer to CRM_Assembly_Instructions.pdf
```

If you see errors such as

```
ERROR: pip's dependency resolver does not currently take into account all
the packages that are installed. This behaviour is the source of
```

these can be ignored.

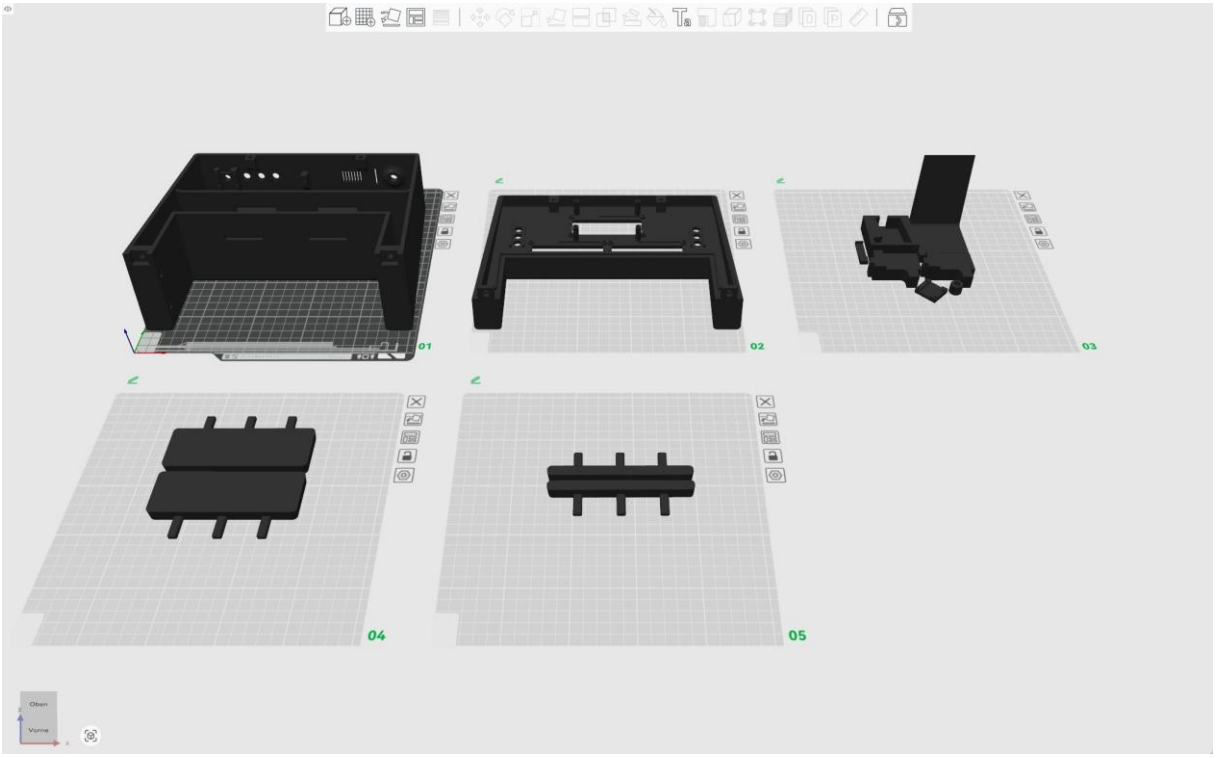
The first installation on a system will not install services so that you can run device tests during assembly (see chapter 6)

If the I2C interface has been activated during installation, a reboot may be requested.

3 Print required Parts

You can now print the parts required for your setup.

Download the Carrera_Race_Monintor.3mf from https://makerworld.com/



You will need to print all of plates 01, 02, 03.

Depending on the track width of your racetrack, you may also need the spacers on plate 04 (Carrera GO, Carrera GO+, Digital 143) or plate 04 (Carrera Profi).

No spacers are required for Carrera Evolution, Carrera Digital 124 and Carrera Digital 132.

4 Prepare Devices

In this step you will prepare the devices required for Carrera Race Monitor.

4.1 LCD Displays

Required Parts:

- 3 LCD displays (#05) which are bundled together with an I2C adapter

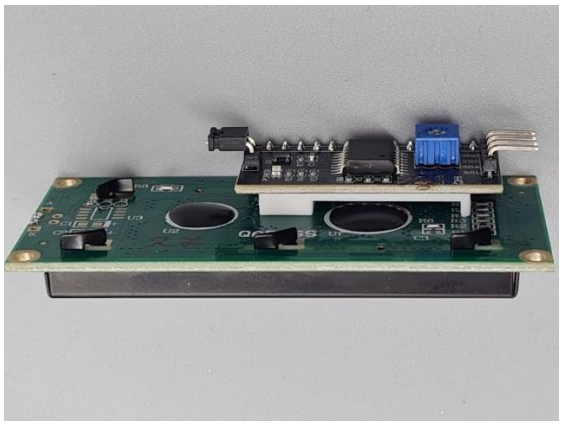
Tools:

- Soldering equipment (#200)
- PCB Clamp (#201)
  Self-printed, such as One of the models from Makerworld
- Fiberglass eraser (#202)

Step 1: Connect I2C adapters to LCD displays

To assure equal spacing without contact, use the spacer **Spacing_I2C_LCD.stl** printed from plate 03.

This will also ensure that you do not need to cut off protruding contacts on the display side.

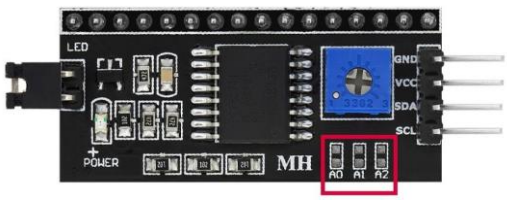


Step 2: Configure individual I2C addresses for each of the LCD displays

Once the I2C adapters have been soldered on the back of the 3 LCD displays, you will need to configure different I2C addresses for each of the displays.

This is done by soldering the pads A0, A1, A2 on the I2C adapters:

- 0x27 : All open
- 0x26 : A0 closed
- 0x25 : A1 closed



Before soldering, clean the pad with a fiberglass eraser (#202) or fine sandpaper and then wipe with isopropyl alcohol.

Note the address on the device.

4.2 Prepare Board for Logic Level Shifter

As the I2C adapter works in the 5V range, whereas GPIO pins only provide 3.3V, a logic level shifter must be used.

The board will allow parallel connection of all 3 LCD displays through JST sockets.

Required Parts:

- Logic Level Converter 3.3V 5V (#06)
- Strip Grid circuit board (#07)

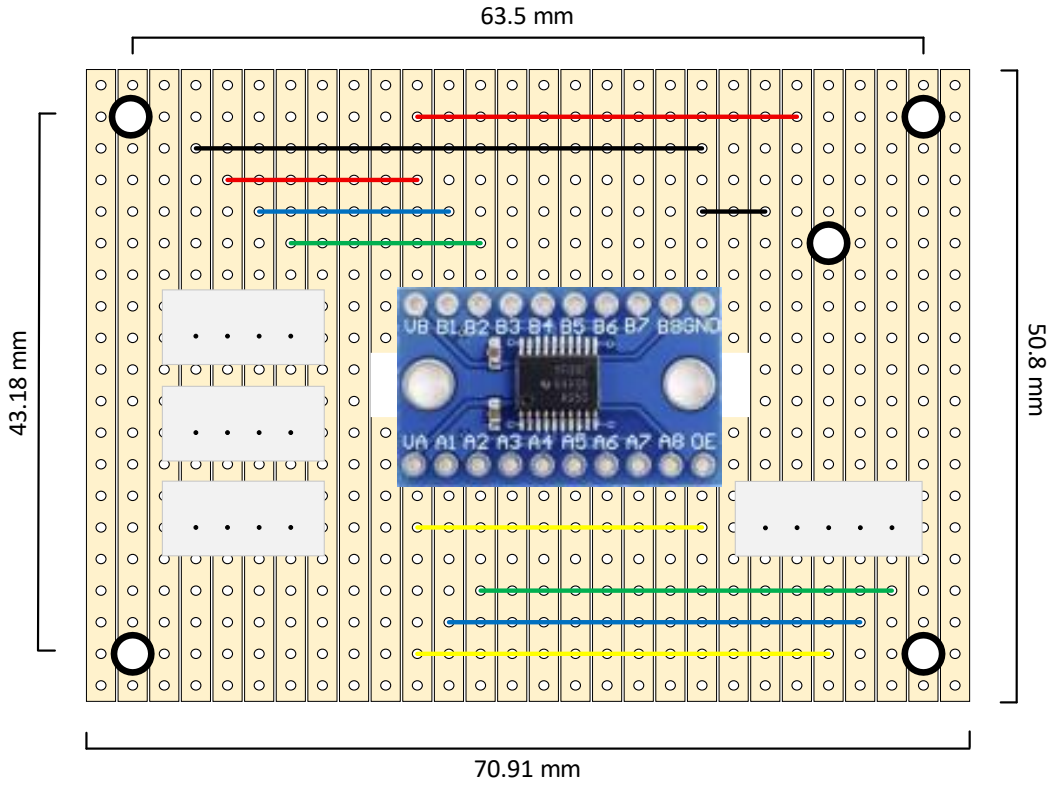
Stock Items:

- 1 JST XH-5 male for circuit board (#100)
- 3 JST XH-4 male for circuit board (#101)
- 24 AWG Solid-Core Wire (black, red, blue, green, yellow) (#102)

Tools:

- Soldering equipment (#200)
- PCP Clamp (#201)
  Self-printed, such as One of the models from Makerworld
- Multimeter (#203)

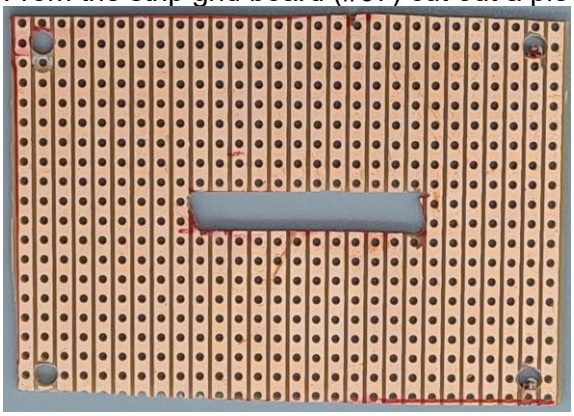
4.2.1 Board Layout



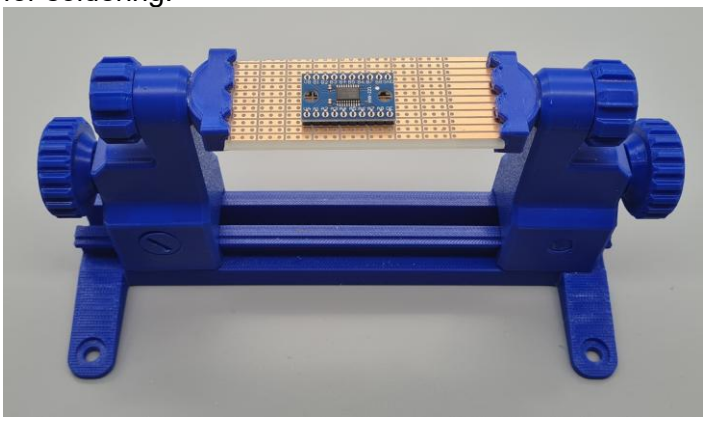
Note that the copper coated side is on the back while the drawing shows it on the front for better transparency.

The additional hole on the right side is for access of the potentiometer of the circuit LCD to allow adjustment of contrast.

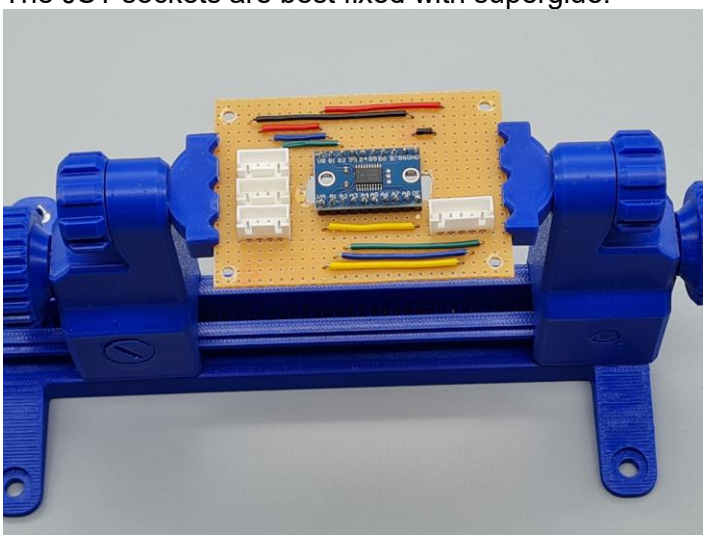
1. From the strip grid board (#07) cut out a piece as shown in the layout, above



2. Drill 3 mm holes at the indicated positions, centered at the grid points.
3. Drill an additional 3 mm hole at the position indicated above.
4. Use a rest piece of a circuit plate to position the 2 10-pin headers and the level shifter for soldering:



5. Place the pre-soldered level shifter, the JST sockets (#100, #101) and the wired connectors (#102) on the non-coated side of the board and fix them.
   The JST sockets are best fixed with superglue.



6. Solder each of the 57 points on the copper-coated side.
7. Check each path for zero resistance using a multimeter (#203).

4.3 Traffic Light Modules

Required Parts:

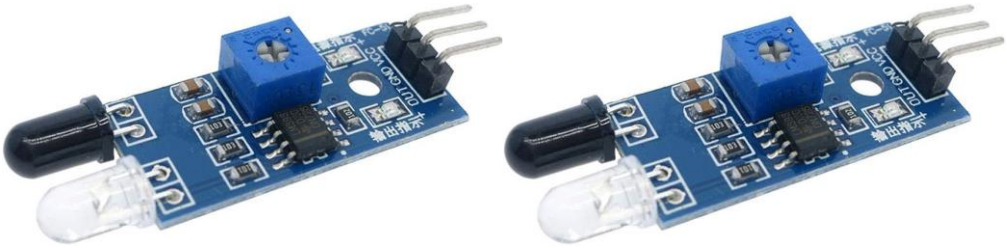
- 2 Traffic Light Modules (#08)



4.4 Obstacle Sensor

Required Parts:

- 2 Obstacle Sensors (#09)



4.5 Push Buttons

Required Parts:

- 3 Push buttons (#10) in colors red, green, yellow



4.6 Rotary Encoder

Required Parts:

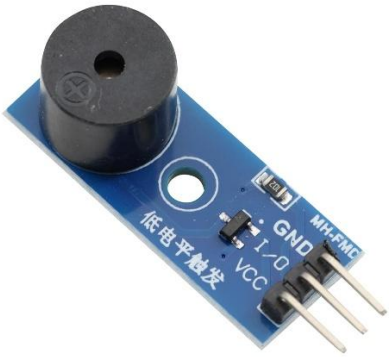
- 1 Rotary Encoder (#11)



4.7 Active Buzzer

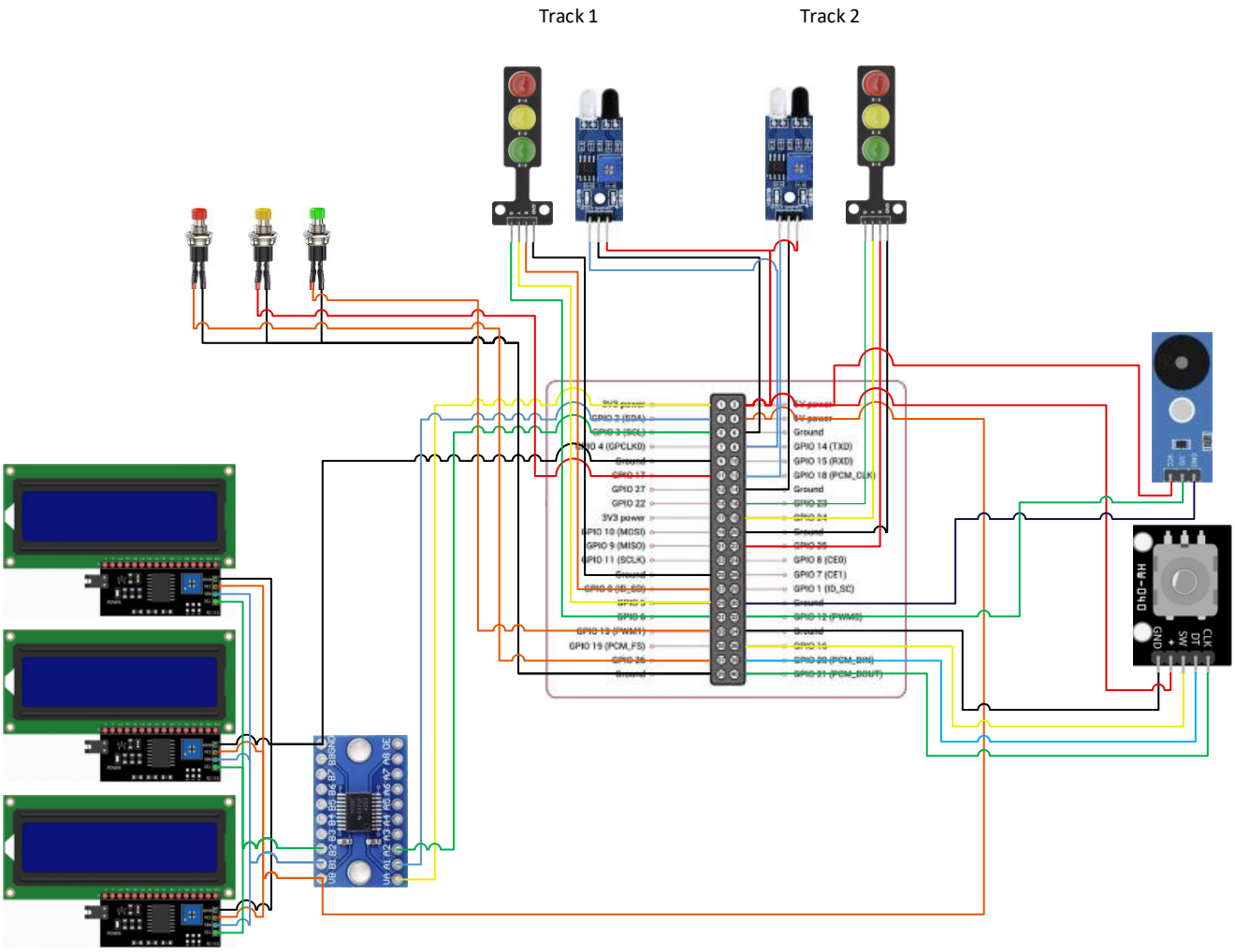
Required Parts:

- 1 Active Buzzer (#12)



5 Prepare Cabling

5.1 Electrical Layout



5.2 Device-GPIO Connections

The following table shows the different devices with their various pins (Pin Out), the GPIO header pins (Pin Board / BCM) to which these need to be connected and the color of the cable to be used (Cable Col).

Device Name	Device Type	Pin Out	Cable Col	Pin Board	BCM	RPI Desc	I2C Address
	Level shifter	GND	black	9		GND	
	Level shifter	VB	red	4		5V Power	
	Level shifter	VA	yellow	1		3V Power	
	Level shifter	A1	blue	3	GPIO 2	SDA	
	Level shifter	A2	green	5	GPIO 3	SCL	
LCD_Track_1	LCD						25
LCD_Track_2	LCD						26
LCD_Circuit	LCD						27
SNS_Track_1	Obstacle_Sensor	VCC	red	2		5V Power	
SNS_Track_1	Obstacle_Sensor	GND	black	6		GND	
SNS_Track_1	Obstacle_Sensor	OUT	blue	8	GPIO 14		
SNS_Track_2	Obstacle_Sensor	VCC	red	2		5V Power	
SNS_Track_2	Obstacle_Sensor	GND	black	14		GND	
SNS_Track_2	Obstacle_Sensor	OUT	blue	12	GPIO 18		
TRL_Track_1	Traffic Light	GND	black	25		GND	
TRL_Track_1	Traffic Light	R	red	27	GPIO 0		
TRL_Track_1	Traffic Light	Y	yellow	29	GPIO 5		
TRL_Track_1	Traffic Light	G	green	31	GPIO 6		
TRL_Track_2	Traffic Light	GND	black	20		GND	
TRL_Track_2	Traffic Light	R	red	22	GPIO 25		
TRL_Track_2	Traffic Light	Y	yellow	18	GPIO 24		
TRL_Track_2	Traffic Light	G	green	16	GPIO 23		
Selector	Rotary Encoder	GND	black	34		GND	
Selector	Rotary Encoder	+	red	2		5V Power	
Selector	Rotary Encoder	SW	yellow	36	GPIO 16		
Selector	Rotary Encoder	DT	blue	38	GPIO 20		
Selector	Rotary Encoder	CLK	green	40	GPIO 21		
Speaker	Passive Buzzer	GND	black	30		GND	
Speaker	Passive Buzzer	I/O	green	32	GPIO 12		
Speaker	Passive Buzzer	VCC	red	2		5V Power	
BTN_Reset	Button	GND	black	39		GND	
BTN_Reset	Button	Signal	red	37	GPIO 26		
BTN_Restart	Button	GND	black	39		GND	
BTN_Restart	Button	Signal	red	11	GPIO 17		
BTN_Start	Button	GND	black	39		GND	
BTN_Start	Button	Signal	red	33	GPIO 13		

5.3 GPIO Header Usage

The following table shows the number of cables (Cnt) connected to each of the GPIO header Pins.

Pin Board	BCM	Function	Cnt
7	GPIO 4	GPCLK0	
10	GPIO 15	RXD	
13	GPIO 27		
15	GPIO 22		
17		3V3 power	
19	GPIO 10	MOSI	
21	GPIO 9	MISO	
23	GPIO 11	SCLK	
24	GPIO 8	CE0	
26	GPIO 7	CE1	
28	GPIO 1	ID_SC	
35	GPIO 19	PCM_FS	
2		5V power	4
39		Ground	3
1		3V3 power	1
3	GPIO 2	SDA	1
4		5V power	1
5	GPIO 3	SCL	1
6		Ground	1
8	GPIO 14	TXD	1
9		Ground	1
11	GPIO 17		1
12	GPIO 18	PCM_CLK	1
14		Ground	1
16	GPIO 23		1
18	GPIO 24		1
20		Ground	1
22	GPIO 25		1
25		Ground	1
27	GPIO 0	ID_SD	1
29	GPIO 5		1
30		Ground	1
31	GPIO 6		1
32	GPIO 12	PWM0	1
33	GPIO 13	PWM 1	1
34		Ground	1
36	GPIO 16		1
37	GPIO 26		1
38	GPIO 20	PCM_DIN	1
40	GPIO 21	PCM_DOUT	1

5.4 Cable Connections

From the 40 pins of the GPIO header, 28 will be used for Carrera Race Monitor.

To make the connections pluggable, the following structure will be applied, in principle:

GPIO Header – 40-Pin cable – JST XH male – JST XH female – cable – Dupont Connector

Required Parts:

- 40-Pin GPIO Rainbow Ribbon Cable (#13)

Stock Items:

- Various JST XH2.54 Sockets and Connectors (#103 … #110)
- 24 AWG Stranded Wire, Silicone (#111)
- Various Dupont Connectors (#112 … 114)
- Cable labels (#115)
- Heat Shrink Tubing (#116)

Tools:

- Crimping Tool (#204)
- Multimeter (#203)
- Soldering Equipment (#200)

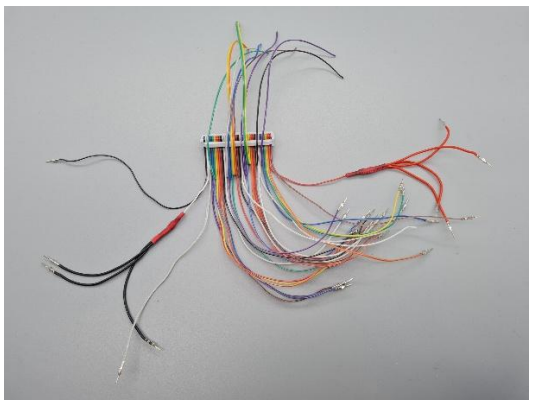
5.4.1 Prepare 40-Pin Rainbow Ribbon Cable

You can easily identify the board number of a pin to which a cable connects by counting from the right:



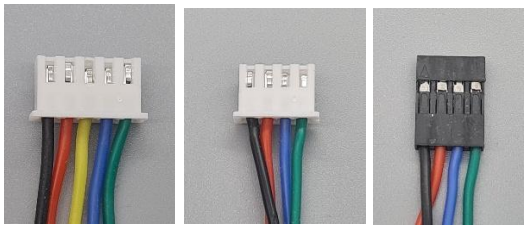
Use the table in chapter 5.3 to identify the required pins/cables

1. Cut the 40-Pin Rainbow Ribbon Cable close to the male connector
2. Solder 4 cable extensions (#111) (red) to cable 2 (5V)
3. Solder 3 cable extensions (#111) (black) to cable 39 (Ground)
4. Protect the solder connections with heat shrink tubing in the respective color
5. Now, isolate the required cables and crimp a JST XH2.54 Pin on the end, including the 5V and Ground extensions



5.4.2 General

Cable connections should follow a consistent sequence of colors/functions, at least on the Raspberry Pi (JST) side:

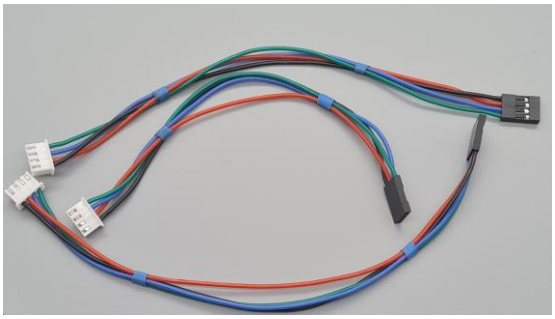


GND(black) - 5V(red) – 3.3V(yellow) – Signal(blue) – Signal (green)

On the device side (Dupont), the sequence depends on the specific device.

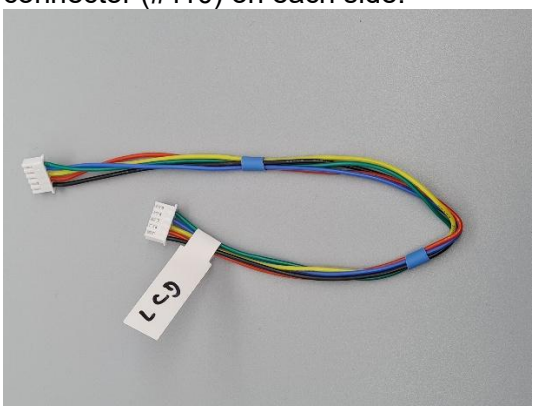
5.4.3 LCD to Level Shifter Board Connection

1. Crimp 3 sets of 4-strand connection cables (#111) of about 15 cm length with a JST XH 4P female connector (#109) on one side and a 4P Dupont Connector (#113) on the other.



5.4.4 LCD to Raspberry Pi Connection

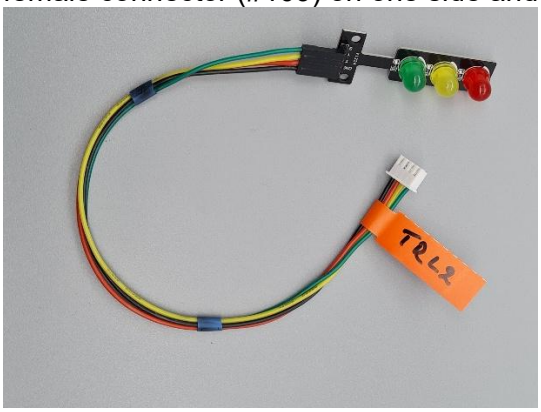
1. Crimp a 5-strand connection cable (#111) of about 20 cm with a JST XH 5P female connector (#110) on each side.



2. Combine cables (9, 4, 1, 3, 4) of the 40-Pin Rainbow Ribbon Cable in a JST XH2.54 5P male socket (#106) so that 9 joins the black cable and 4 the green cable of the female connector.

5.4.5 Traffic Light Connection

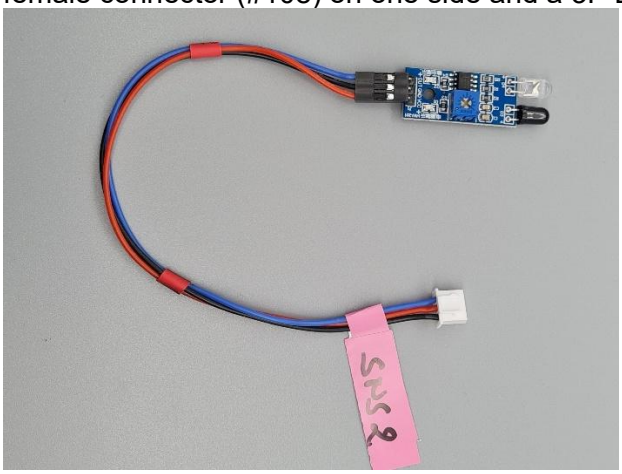
1. Crimp 2 sets of 4-strand connection cables (#111) of about 20 cm with a JST XH 4P female connector (#109) on one side and a 4P Dupont connector (#113) on the other



2. For Traffic Light 1, combine cables (25, 27, 29, 31) of the 40-Pin Rainbow Ribbon Cable in a JST XH2.54 4P male socket (#105) so that 25 joins the black cable and 31 the green cable of the female connector.
3. For Traffic Light 2, combine cables (20, 22, 18, 16) of the 40-Pin Rainbow Ribbon Cable in a JST XH2.54 4P male socket (#105) so that 20 joins the black cable and 16 the green cable of the female connector.

5.4.6 Obstacle Sensor Connection

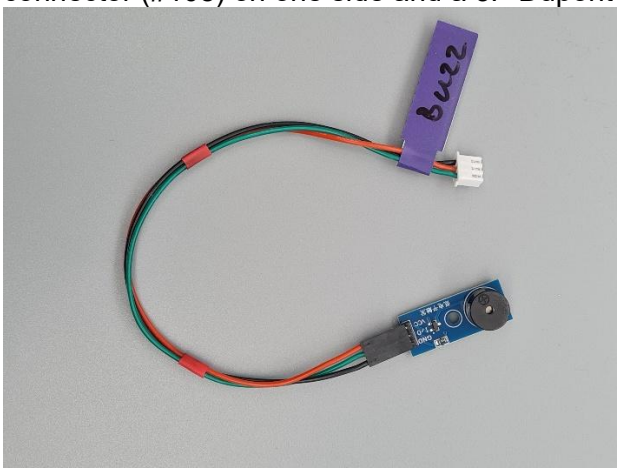
1. Crimp 2 sets of 3-strand connection cables (#111) of about 20 cm with a JST XH 3P female connector (#108) on one side and a 3P Dupont connector (#112) on the other



2. For Sensor 1, combine cables (6, 2, 8) of the 40-Pin Rainbow Ribbon Cable in a JST XH2.54 3P male socket (#104) so that 6 joins the black cable and 8 the blue cable of the female connector.
   NOTE: that the sequence of colors is different on the JST and Dupont side
3. For Sensor 2, combine cables (14, 2, 12) of the 40-Pin Rainbow Ribbon Cable in a JST XH2.54 3P male socket (#104) so that 14 joins the black cable and 12 the blue cable of the female connector. .
   NOTE: that the sequence of colors is different on the JST and Dupont side

5.4.7 Buzzer Connection

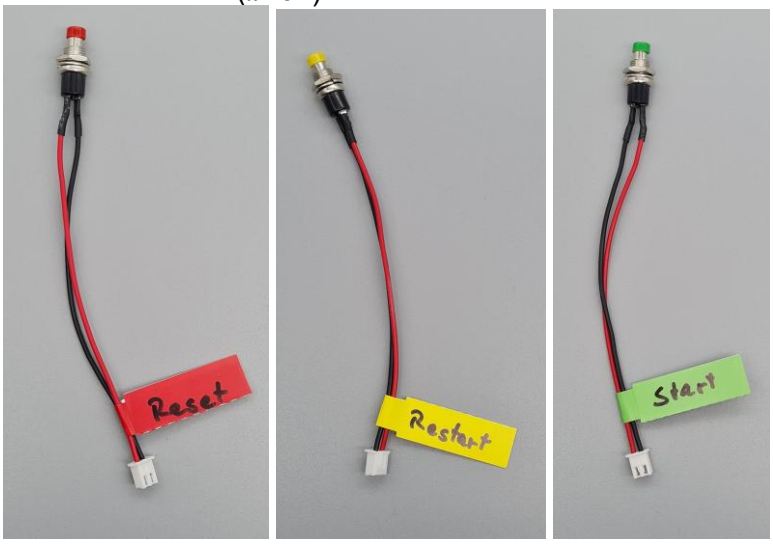
1. Crimp a 3-strand connection cable (#111) of about 20 cm with a JST XH 3P female connector (#108) on one side and a 3P Dupont connector (#112) on the other



2. Combine cables (30, 2, 32) of the 40-Pin Rainbow Ribbon Cable in a JST XH2.54 3P male socket (#104) so that 30 joins the black cable and 32 the green cable of the female connector.
   NOTE: that the sequence of colors is different on the JST and Dupont side

5.4.8 Push Button Connections

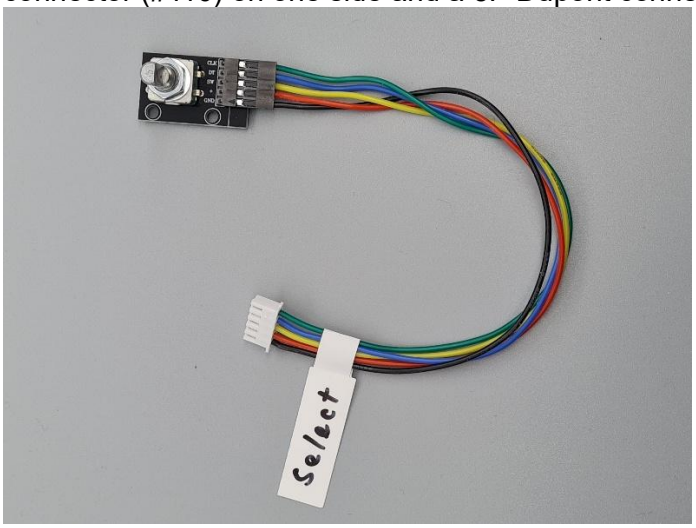
1. For each of the 3 Push Buttons (#10), crimp the available cables with a JST XH 2P female connector (#107)



2. For the Reset Button, combine cables (39, 37) of the 40-Pin Rainbow Ribbon Cable in a JST XH2.54 2P male socket (#103) so that 39 joins the black cable and 37 the red cable of the female connector.
3. For the Restart Button, combine cables (39, 37) of the 40-Pin Rainbow Ribbon Cable in a JST XH2.54 2P male socket (#103) so that 39 joins the black cable and 37 the red cable of the female connector.
4. For the Start Button, combine cables (39, 33) of the 40-Pin Rainbow Ribbon Cable in a JST XH2.54 2P male socket (#103) so that 39 joins the black cable and 37 the red cable of the female connector.

5.4.9 Rotary Encoder Connection

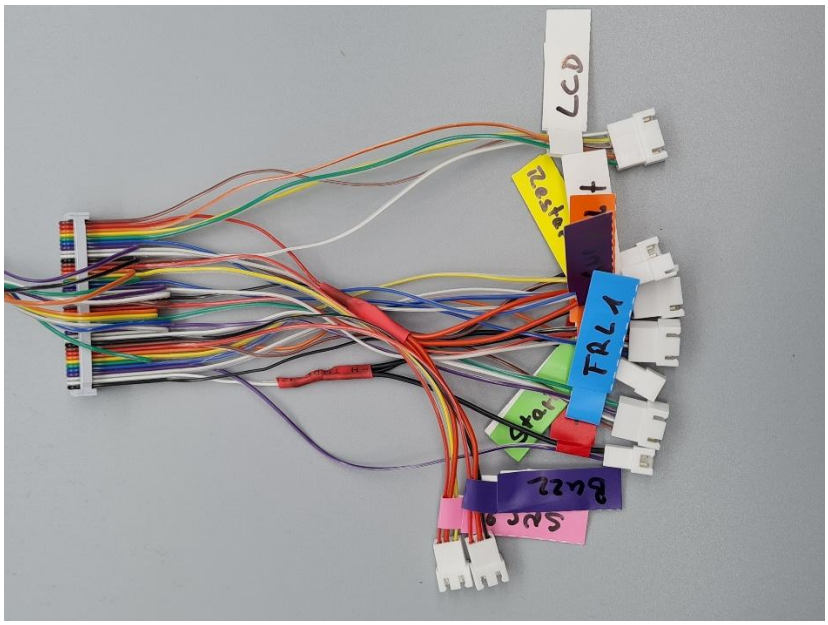
1. Crimp a 5-strand connection cable (#111) of about 20 cm with a JST XH 5P female connector (#110) on one side and a 5P Dupont connector female (#114) on the other



2. Combine cables (34, 2, 36, 38, 40) of the 40-Pin Rainbow Ribbon Cable in a JST XH2.54 5P male socket (#106) so that 34 joins the black cable and 40 the green cable of the female connector.

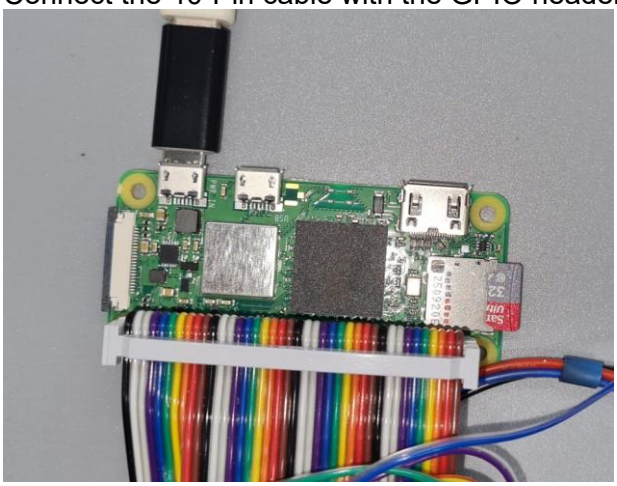
5.4.10 Final 40-Pin Rainbow Cable

Finally, your 40-Pin cable will look like this:



6 Test Connections

1. Connect the 40-Pin cable with the GPIO header of the Raspberry Pi



2. Connect all devices with their corresponding sockets
3. Power up the Raspberry Pi
4. Connect through SSH
   ```
   ssh <user>@<host>
   ```
5. Check that the LCD displays use the correct I2C addresses:
   ```
   sudo i2cdetect -y 1
   ```
   You should see something like this:
   ```
        0  1  2  3  4  5  6  7  8  9  a  b  c  d  e  f
   00:                         -- -- -- -- -- -- -- --
   10: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
   20: -- -- -- -- -- 25 26 27 -- -- -- -- -- -- -- --
   30: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
   40: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
   50: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
   60: -- -- -- -- -- -- -- -- -- -- -- -- -- -- -- --
   70: -- -- -- -- -- -- -- --
   ```
   If you do not see this result, you need to check the LCD devices for correct connections and correct address setting.
   Probably, you may want to check each device individually using a breadboard.
6. Activate the project's virtual environment
   ```
   cd prg/Carrera_Race_Monitor
   source .venv/bin/activate
   ```
7. Run the Device Test
   ```
   python code/crm_device_manager.py
   ```

The device manager (`crm_device_manager.py`) ,when started individually, will run an infinite test sequence which can be stopped with <Ctrl+C>

```
(.venv) sn@raspi12:~/prg/Carrera_Race_Monitor $ python code/crm_device_manager.py
Reset button initialized.
Restart button initialized.
Start button initialized.
Selector initialized.
Selector button initialized.
Track 1 sensor initialized.
Track 1 occupied: False.
Track 2 sensor initialized.
Track 2 occupied: False.
Traffic light for Track 1 initialized.
Traffic light for Track 2 initialized.
Starting Traffic Light Cycle: Red, Yellow, Green, Off
Speaker Info:
Speaker max_tone: A5
Speaker min_tone: A3
Speaker mid_tone: A4
Speaker octaves: 1

Starting Speaker Cycle: Fanfare
LCD for Track 1 initialized.
LCD for Track 2 initialized.
LCD for Circuit initialized.
2026-10-03 18:17:02.231256: Reset button pressed.
2026-10-03 18:17:06.186109: Restart button pressed.
2026-10-03 18:17:09.968227: Start button pressed.
2026-10-03 18:17:22.953661: Selector rotated clockwise.
2026-10-03 18:17:24.160610: Selector rotated counter-clockwise.
2026-10-03 18:17:25.511175: Selector button pressed.
2026-10-03 18:17:39.461936: Track 1 car detected.
2026-10-03 18:17:39.467195: Track 1 occupied: True.
2026-10-03 18:17:40.319291: Track 1 car cleared.
2026-10-03 18:17:40.325691: Track 1 occupied: False.
2026-10-03 18:17:48.854758: Track 2 car detected.
2026-10-03 18:17:48.865090: Track 2 occupied: True.
2026-10-03 18:17:49.896017: Track 2 car cleared.
2026-10-03 18:17:49.901265: Track 2 occupied: False.
^CExiting...
```

You will need to do the following tests:

- Check the LCD displays:
  They should display their identity (Track 1, Track 2, Circuit) together with some additional text. Also, the background light will be switched on and off.
  You may need to adjust the contrast using the potentiometer of the I2C adapter.
- Check Traffic Lights:
  In a sequence, there should be red, red+yellow, green, off and all blinking.
  If one of the colors does not show up brightly, you may need to check cable connections.
- Check Buzzer:
  The buzzer will play two sequences (fanfare, alarm) with 5 sec pause.
  If you do not hear the sound you may need to check the buzzer connections
- Test Track sensors:
  Normally, the visible range of the sensors should be free. Then, after the sensor is initialized, the current status will be logged (Track x occupied: False).
  Now you can block the sensor which will be indicated by the sensor's status LED.
  At the same time, you should see a log output "Track x car detected" and "Track x occupied: True"
  When you release the block, this should also result in a corresponding log output.
  If you do not see the log output, you may need to check the cable connections of the failing sensor.
- Test Push Buttons:
  Press each button: Red (Reset), Yellow (Restart), Green (Start).
  Each button press should issue a log output indicating the button type.
  If the log output is missing or the logged button type is wrong, you may need to check the buttons cable connection.
- Test Rotary Encoder:
  Turn the encoder button to the left and to the right and push it.
  You should see corresponding log output.
  If you do not see the log output, you may need to check the cable connection.

7 Assemble the System

7.1 Press Threaded Inserts

Tools:

- Heat Insert Press (#205)

7.1.1 Top Connectors

Stock Items:

- 2 Threaded Inserts M4x8x6 (#150)
- 2 Cylinder Screws M4x16MM (#151)

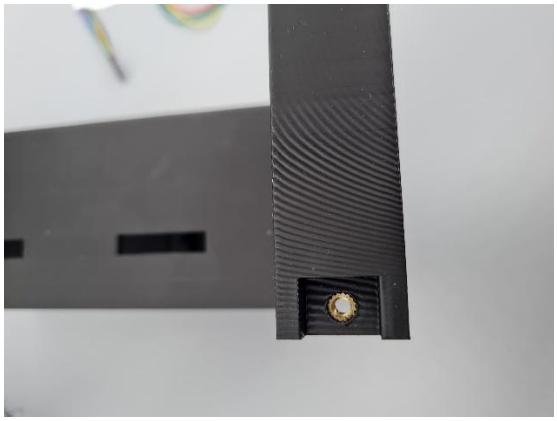
Use printed item "Back_Stand.stl" from plate 3 for exact vertical orientation of the back part when pressing the insert.



7.1.2 Bottom Connectors

Stock Items:

- 2 Threaded Inserts M4x4x6 (#152)
- 2 Cylinder Screws M4x8MM (#153)



7.1.3 Camera Adapter

The Carrera Race Monitor features an optional Raspberry Pi Camera which can be attached to the top of the housing.

Stock Items:

- Threaded Insert 1/4"x6.4x8 (#155)



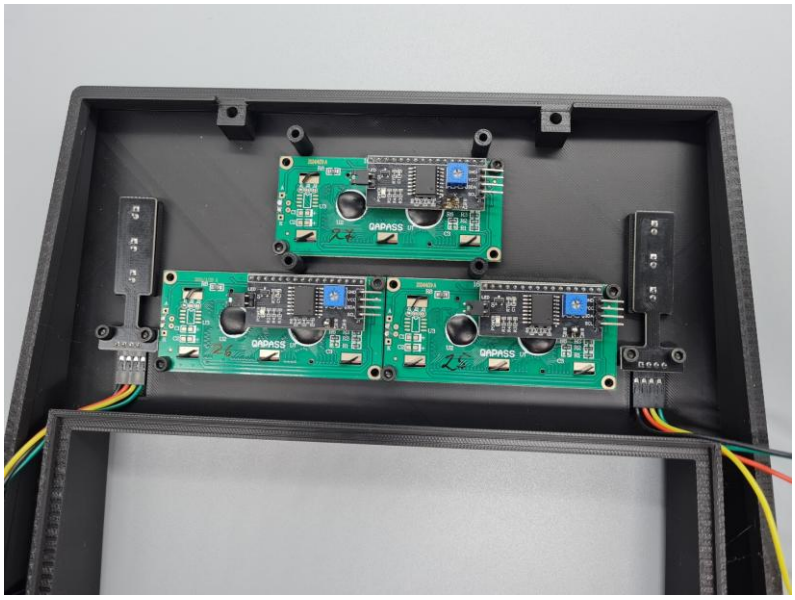
7.2 Assemble Front Part

Stock Items:

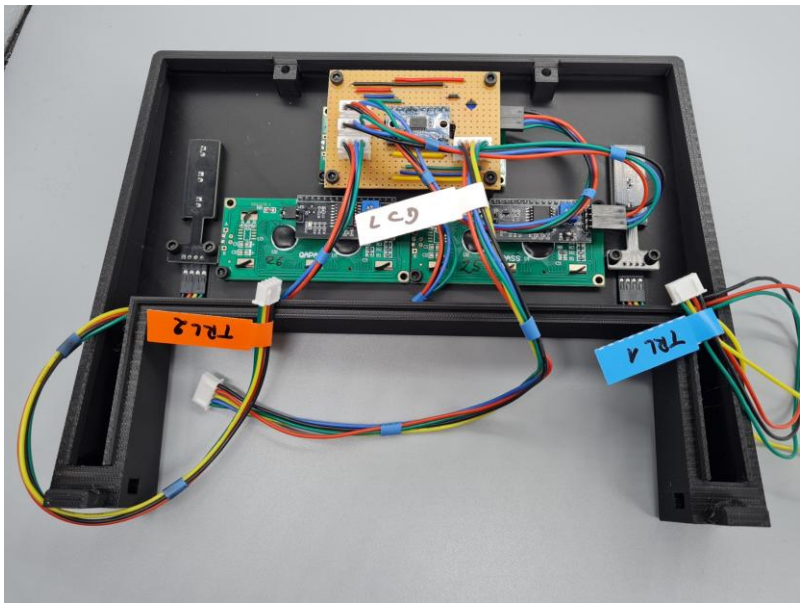
- 14 Cylinder Screws M3x6MM (#154)

It is recommended mounting the traffic lights with cables already connected.

Assure correct positioning of displays according to their I2C address (25, 26, 27) as shown below.



Finally, mount the Level Shifter Board and connect the LCD displays.



7.3 Assemble Back Part

7.3.1 USB Cable

The USB cable will allow powering the Raspberry Pi from outside the housing.

The outside contact is USB-C and an adapter will be used to connect to the Pi's Micro USB.

Required Parts:

- USB C Cable (#14)

Tools:

- PH1 Screwdriver with long shaft (#206)




Fixing the screws is a little bit tricky. It may be helpful to glue the screw on the screwdriver to facilitate placement. Also, if the bottom screw cannot be fixed, it is sufficient to fix the upper screw.

7.3.2 Rotary Encoder

Required Parts

- Rotary Encoder (#11)

Stock Items

- 2 Cylinder Screws M3x6MM (#154)

Tools:

- Hex Key Allen Wrench 3.5 mm (#207)
- Socket Wrench 10 mm (#208)




7.3.3 Push Buttons

Required Parts:

- 1 Push Button Red (#10)
- 1 Push Button Yellow (#10)
- 1 Push Button Green (#10)

See chapter 5.4.8

Tools:

- Metal Drill Bit 8.0 mm (#209)
- Socket Wrench 10 mm (#208)

If the holes for the bush buttons turn out to be a little bit narrow, you can gently widen them with an 8 mm Metal Drill Bit.



Now, attach connection cable for Rotary Encoder:



7.3.4 Buzzer

Required Parts:

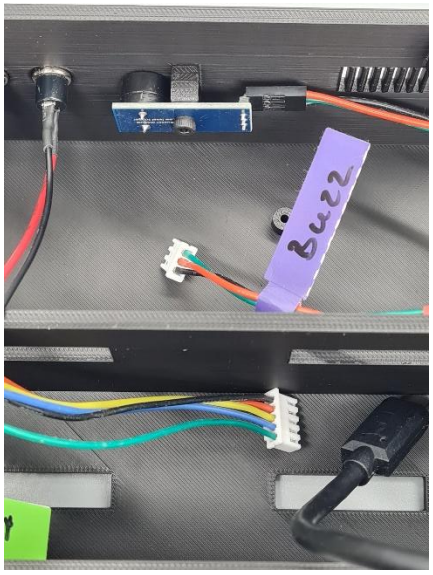
- Buzzer (#12)

Stock Items:

- 1 Cylinder Screw M3x6MM (#154)

Tools:

- Hex Key Allen Wrench 3.5 mm (#207)



7.3.5 Mount Raspberry Pi

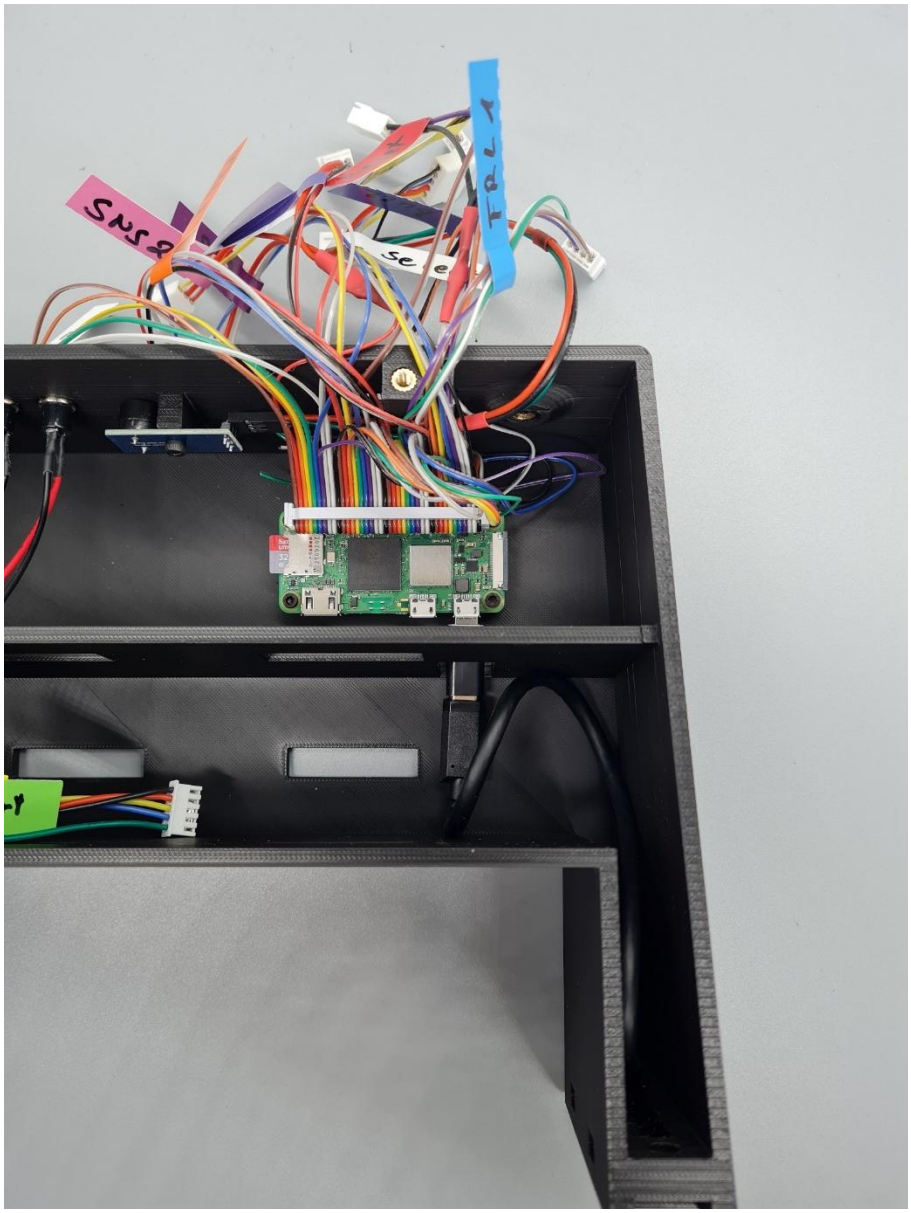
Required Parts:

- USB-C to Micro USB Adapter

Stock Items:

- 4 Cylinder Screw M2.5x8 (#157)

1. First, plug the USB-C-to-Micro-USB adapter onto the USB cable.
2. Then, feed the cable through the right opening of the middle level and plug the adapter into the Raspberry Pi's power socket outside the housing.
3. Now push the Raspberry Pi gently in place and fix it with 4 screws.



7.3.6 Preparing the Obstacle Sensors

Printed Parts:

- OD_Mount_1.stl (plate 3)
- OD_Mount_2.stl (plate 3)
- OD_Mount_Stand.stl (plate 3)

Parts:

- 2 Obstacle Sensors (#09)

Stock Items:

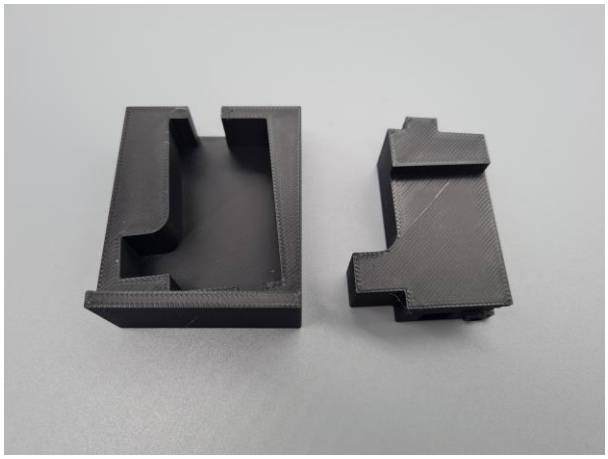
- 2 Threaded Inserts M4x6x6 (#158)
- 2 Cylinder Screws M4x8MM (#153)
- 2 Washers M4 x 12 mm (#159)
- 2 Cylinder Screws M3x6MM (#154)

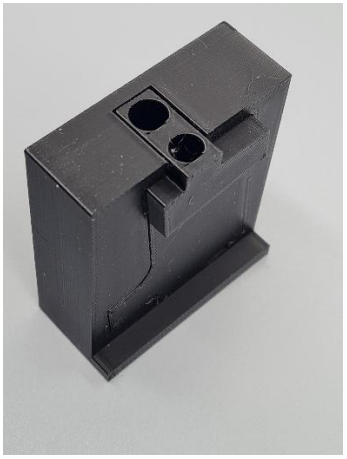
Tools:

- Heat Insert Press (#205)

7.3.6.1 Press Threaded Inserts

The OD_Mount_Stand.stl will assure exact vertical orientation of the OD_Mount_x.stl parts:





The threaded inserts must be inserted in the smaller right hole.

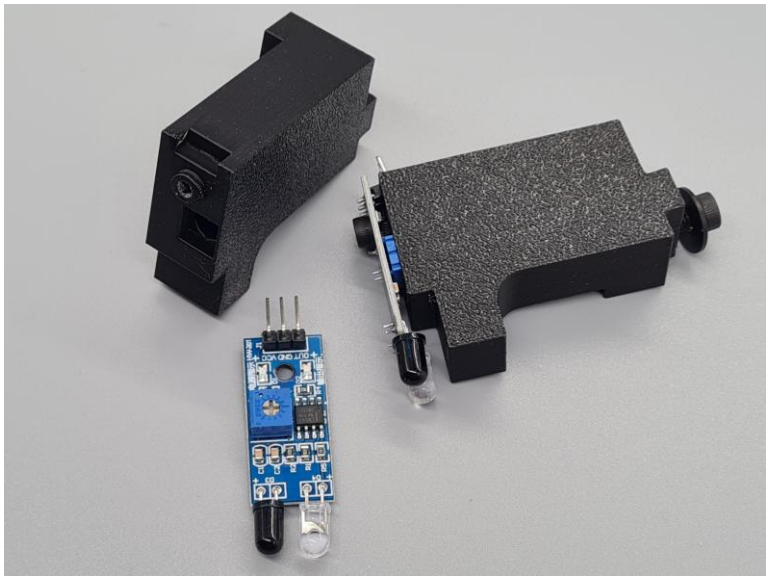
The larger left hole is for calibration of the obstacle sensors.



7.3.6.2 Mount Obstacle Sensors

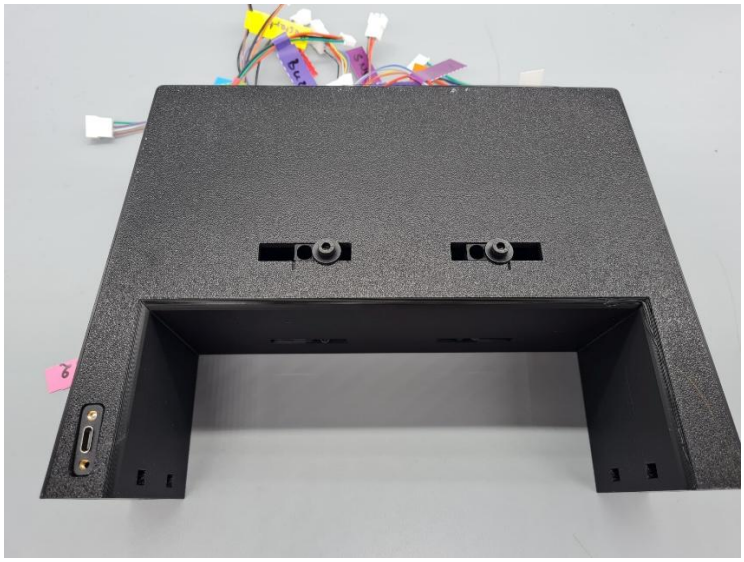
The potentiometer of the obstacle sensors fits exactly in the square hole of the support and can be reached through the hole from outside for calibration.

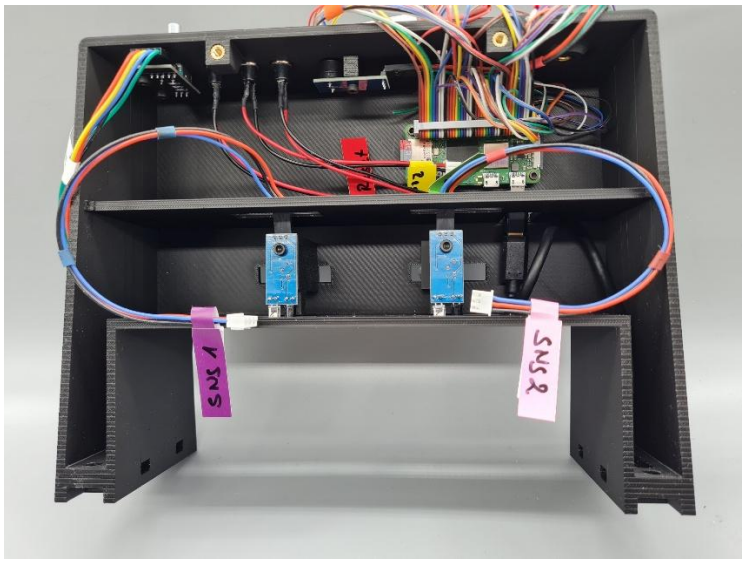
The obstacle sensors are fixed with a M3x6MM Cylinder Screw (#154)



7.3.7 Mounting Obstacle Sensors

With the hold-down screws removed, insert the obstacle sensor supports into the main housing and tighten with washers and screws:





Choose the sensor position according to your track width:

- Outer position: 100 mm
  (Evolution, Digital 132, Digital 124)
- Marked position as in image above: 71 mm
  (Profi)
- Inner position: 60 mm
  (GO!!!, GO+, Digital 143)

7.4 Connect and Test

1. Close all associated connections between devices and Raspberry Pi while the housing is still open.
2. Plug in a power cable
3. Run the same test as in chapter 6

7.5 Close Housing

Printed Parts:

- RE_Button.stl (Plate 3)

Stock Items:

- 2 Cylinder Screws M4x16MM (#151)
- 2 Cylinder Screws M4x8 MM (#153)

1. Stow all cables in the upper compartment.
2. Close the housing.
   Make sure that no cables get pinched when closing the case.
3. Fix bottom connectors with two M4x8MM screws.
4. Fix upper connectors with two M4x16MM screws.
5. Push the printed button for the rotary encoder onto its axle.
   You need to make sure that the cut-out section of the button aligns with the corresponding section of the axle.

7.6 Final Test

To make sure that closing has not corrupted any of the connections, repeat the test described in chapter 6.







8 Adapting the Configuration

If you have chosen to use other I2C addresses or other GPIO contacts than suggested in this document, you will need to adjust the configuration file.

You will find it under

`~/prg/Carrera_Race_Monitor/config/crm_config.json`

```
{
    "config": {
        "lng": "en",
        "tracks": {
            "Track_1": {
                "enabled": true,
                "handicap": 0.0
            },
            "Track_2": {
                "enabled": true,
                "handicap": 0.0
            }
        },
        "start_control": false,
        "race_mode": "OPEN",
        "race_laps": 10,
        "race_time_min": 5,
        "play_sound": true
    },
    "devices": {
       "LCD_Track_1": {
            "type": "LCD",
            "params": {
                "I2C_address": "0x25"
            }
        },
        "LCD_Track_2": {
            "type": "LCD",
            "params": {
                "I2C_address": "0x26"
            }
        },
        "LCD_Circuit": {
            "type": "LCD",
            "params": {
                "I2C_address": "0x27"
            }
        },
        "SNS_Track_1": {
            "type": "Obstacle_Sensor",
            "params": {
                "GPIO_pin": 14,
                "bounce_time": 0.05
            }
        },
        "SNS_Track_2": {
            "type": "Obstacle_Sensor",
            "params": {
                "GPIO_pin": 18,
                "bounce_time": 0.05
            }
        },
```

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        "TRL_Track_1": {
            "type": "Traffic_Light",
            "params": {
                "GPIO_pin_red": 0,
                "GPIO_pin_yellow": 5,
                "GPIO_pin_green": 6
            }
        },
        "TRL_Track_2": {
            "type": "Traffic_Light",
            "params": {
                "GPIO_pin_red": 25,
                "GPIO_pin_yellow": 24,
                "GPIO_pin_green": 23
            }
        },
        "Selector": {
            "type": "Rotary_Encoder",
            "params": {
                "GPIO_pin_CLK": 21,
                "GPIO_pin_DT": 20,
                "GPIO_pin_SW": 16
            }
        },
        "Speaker": {
            "type": "Passive_Buzzer",
            "params": {
                "GPIO_pin": 12
            }
        },
        "BTN_Reset": {
            "type": "Button",
            "params": {
                "GPIO_pin": 26,
                "bounce_time": 0.1
            }
        },
        "BTN_Restart": {
            "type": "Button",
            "params": {
                "GPIO_pin": 17
            }
        },
        "BTN_Start": {
            "type": "Button",
            "params": {
                "GPIO_pin": 13
            }
        }
    }
}
```